

LESSON 8-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Multiplication Equations**

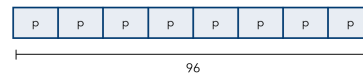
Maryland Standard 6.EE.B.7 • Learning target – I can solve a multiplication equation by dividing both sides by the same number.

REMEMBER

- In $6x$, the **coefficient** 6 means $6 \times x$.
- To solve $ax = b$, **divide both sides by** a . Division undoes multiplication.
- **Check:** the coefficient times your solution should equal the right side.

WORKED EXAMPLE

Solve $8p = 96$.



Step 1 Divide both sides by 8: $p = 96 \div 8 = 12$

Step 2 Check: $8 \times 12 = 96$, so it checks.

Answer $p = 12$

WORD BANK

coefficient – the number times the variable

equal – the same amount

divide both sides – undo the times

PRACTICE

show your work • label units

1. Solve each equation.

a. $4x = 36$ _____

b. $7n = 56$ _____

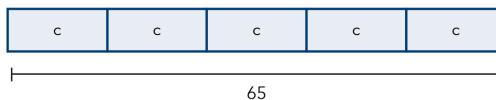
c. $9y = 108$ _____

d. $12k = 84$ _____

e. $5m = 85$ _____

f. $3w = 4.5$ _____

2. Write a multiplication equation for the tape diagram. Then solve it.



Equation and solution _____

3. **Bus Fare.** A family pays \$11.25 for 5 bus fares. Each fare costs the **same**. Write and solve an equation for f , the cost of one fare. Check your answer.

Step 1 $5 \times \text{one fare} = \text{total}$.

Step 2 Divide both sides by 5.

Show your work.

Answer _____ dollars

- 4. Garden Rows.** Ms. Bell plants 84 seedlings in 6 equal rows. Write and solve a multiplication equation for s , the seedlings in each row.

Step 1 Rows $\times$ seedlings in each row = all seedlings.

Step 2 Divide both sides by 6.

Answer _____ seedlings

- 5. Compare.** Without solving, tell which equation has the greater solution: $4n = 80$ or $5n = 80$. Explain. Then solve both to check.

Frame ____ has the greater solution because the same total split into ____ groups makes each group ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Find a package with equal parts (eggs, juice boxes, crayons). Write a multiplication equation for it and solve for one part.

LESSON 8-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Division Equations**

Maryland Standard 6.EE.B.7 • Learning target – I can solve a division equation by multiplying both sides by the same number.

REMEMBER

- To solve $x \div a = b$, **multiply both sides by a** .
- The solution is the **whole**, so it is greater than b .
- **Check:** divide your solution by a .

WORKED EXAMPLE

Solve $n \div 4 = 15$.

Step 1 Multiply both sides by 4: $n = 15 \times 4 = 60$

Step 2 Check: $60 \div 4 = 15$, so it checks.

Answer $n = 60$

WORD BANK

divisor – the number you divide by

share equally – split into equal groups

multiply both sides – undo the divide

PRACTICE

show your work • label units

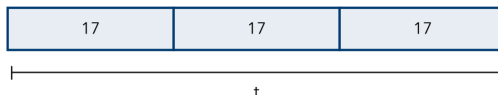
1. Solve each equation.

a. $n \div 7 = 6$ _____

b. $y \div 5 = 13$ _____

c. $k \div 10 = 4.5$ _____

2. Write a division equation for the tape diagram. Then solve it.



Equation and solution _____

3. **Jump Rope.** A rope is cut into 8 **equal** pieces. Each piece is 2.5 meters long. Write and solve a division equation for r , the whole rope.

Step 1 Whole rope $\div 8 =$ one piece.

Step 2 Multiply both sides by 8.

Show your work.

Answer _____ meters

4. **Pizza Party.** A pizza is cut into equal slices. 6 friends share it, and each gets 2 slices. Write and solve a division equation for z , the slices in the pizza.

Step 1 Slices $\div$ friends = slices for each friend.

Step 2 Multiply both sides by 6.

Answer _____ slices

- 5. Find the Error.** Kayla and 3 friends share cards equally. Each gets 9 cards. Kayla writes $c \div 3 = 9$. Explain her mistake. Write and solve a correct equation.

Frame Kayla's mistake is _____. The cards are shared by _____ people, so $c =$ _____.

TEST PRACTICE

choose the best answer

- 6.** Solve $y \div 6 = 9$.

A $y = 1.5$

B $y = 3$

C $y = 15$

D $y = 54$

- 7.** 12 students share p pretzels equally. Each gets 5. Which equation fits?

A $p \div 12 = 5$

B $12 \div p = 5$

C $5p = 12$

D $p - 12 = 5$

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____